

INTERVIEW PREPARATION — PROJECT QUESTION BANK

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Files in this Folder

File	Content	Questions
01_GeoFenced_Firearm_Safety_Interview.md	Smart Geo-Fenced Firearm Safety System	50 Q&A
02_AIxAI_Negotiation_Interview.md	AIxAI Negotiation System	50 Q&A
03_Agentic_AI_CustomerSupport_Interview.md	Agentic AI Customer Support (Internship)	50 Q&A
04_Cross_Project_And_RapidFire.md	Cross-Project (20 Q) + Rapid-Fire (30 Q)	50 Q&A

Total: 200 Questions

Structure of Each Project File

Each of the 3 project files follows this structure:

A. Project Understanding	– Q1–Q5
B. Architecture & System Design	– Q6–Q12
C. Technology Deep Dive	– Q13–Q20
D. Code & Implementation	– Q21–Q25
E. Database	– Q26–Q30
F. API & Backend	– Q31–Q34
G. Security	– Q35–Q38
H. Failure & Edge Cases	– Q39–Q42
I. Improvement & Scalability	– Q43–Q46

J. Interviewer Pressure — Q47-Q50

- + Most Important 10 Questions
 - + Questions I Must Never Get Wrong
 - + 5-Minute Project Explanation
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Quick Revision Checklist

Project 1 — Geo-Fenced Firearm Safety System

- ☐ Can explain the full detect → verify → decide → lock flow
- ☐ Knows why edge computing over cloud decision-making
- ☐ Can explain RFID and how it works
- ☐ Knows ESP32 PWM servo control basics
- ☐ Can explain the authorized ID local storage & sync mechanism
- ☐ Knows the fail-safe design (default = locked on power loss)
- ☐ Has memorized the 5-minute project pitch

Project 2 — AIxAI Negotiation System

- ☐ Can explain the 6-phase negotiation workflow
- ☐ Can derive the utility calculation formula from memory
- ☐ Can defend why this is "AI" (utility-based agents)
- ☐ Knows that "reinforcement" is NOT reinforcement learning
- ☐ Can explain churn prediction and its features
- ☐ Knows FastAPI vs Flask vs Django trade-offs
- ☐ Has memorized the 5-minute project pitch

Internship — Agentic AI Customer Support

- ☐ Can explain RAG end-to-end (chunk → embed → store → retrieve → generate)
- ☐ Knows what is stored in ChromaDB (text + embedding + metadata)

- [] Can explain semantic search and why it beats keyword search
 - [] Knows why LangGraph over plain Python if-else
 - [] Can explain prompt injection and mitigations
 - [] Knows ChromaDB vs FAISS trade-offs
 - [] Has memorized the 5-minute project pitch
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Key Interview Principles

1. **Own your projects.** Speak with confidence. You built these.
2. **Be precise about AI terms.** Know the difference between: utility-based AI vs RL vs LLM, RAG vs fine-tuning, LangGraph vs LangChain Agents.
3. **Never bluff.** If a detail is unspecified, say so and give the correct theoretical answer.
4. **Defend your design decisions.** Know WHY you chose each technology.
5. **Know the failure modes.** Every interviewer will ask "what if X fails?"
6. **Practice speaking, not reading.** Record yourself giving the 5-minute explanations.